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# 3D-Aware VLMs with Implicit and Explicit Geometries

*VLM-IE3D Equips Vision-Language Models with Two Complementary Geometric Representations Derived Purely from RGB Video, Setting New State-of-the-Art on Four 3D Spatial Benchmarks*

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Vision-language models have demonstrated impressive language and 2D visual understanding but struggle when tasks demand precise 3D spatial reasoning. A new ECCV 2026 paper from Nanyang Technological University introduces **VLM-IE3D**, which equips standard VLMs with two complementary 3D geometric representations—implicit and explicit—derived from RGB video alone, with no depth sensors or 3D supervision. The framework achieves state-of-the-art performance on 3D video detection, 3D visual grounding, 3D dense captioning, and spatial reasoning benchmarks.

## AT A GLANCE

**Problem:** VLMs built on 2D visual pretraining lack the fine-grained 3D spatial understanding needed for grounding, detection, and spatial reasoning in 3D scenes.

**Why it matters:** 3D-capable VLMs are essential for robotics, AR/VR, and embodied AI, yet requiring depth sensors limits real-world deployment.

**Method:** Implicit Geometry Tokens (coarse global priors) and Explicit Geometry Tokens (fine-grained reconstructed structure) fused with 2D visual tokens via a 3D-aware adapter.

**Result:** State-of-the-art on ScanQA (+6.4 pp), ScanRefer (+7.7 pp), Scan2Cap (+8.5 CIDEr), and EmbodiedScan (+4.7 mAP).

**Evidence:** RGB-only at inference; ablations confirm both geometry types are necessary and complementary.

## The Big Picture

Modern VLMs are trained on billions of image-text pairs, giving them powerful 2D visual understanding and broad world knowledge. Yet 3D spatial tasks—locating objects relative to each other in 3D space, predicting 3D bounding boxes, or answering questions about scene geometry—

remain a persistent weakness. These tasks require a model to reason about depth, surface orientation, and object extent that is only partially recoverable from 2D images.

Existing solutions typically assume access to depth sensors or LiDAR at inference time, which is unrealistic for camera-only deployments. VLM-IE3D bridges this gap: starting from ordinary RGB video, it derives two complementary geometric representations and injects them into the VLM as geometric token streams.

## Why Existing Approaches Fall Short

Three prior strategies have been tried with limited success:

**Monocular depth augmentation:** Estimated depth maps are fed alongside RGB images. Depth estimation errors propagate to downstream tasks, and estimated depth lacks surface-normal precision required for grounding.

**Implicit-only geometry:** Some recent VLMs inject geometric priors learned from video pretraining via contrastive objectives. These provide coarse global layout but lack the local precision needed for fine-grained 3D grounding—a deficiency VLM-IE3D directly addresses with explicit geometry tokens.

**Explicit-only geometry:** Sparse point clouds from structure-from-motion (SfM) encode precise local structure but lack the holistic scene layout captured by implicit priors. Without the global context, reasoning about scene-level spatial relationships degrades.

VLM-IE3D combines both: implicit for global layout, explicit for local precision.

## Core Mechanism

**Implicit Geometry Tokens (IGTs)** are produced by a lightweight video encoder that processes  $T$  frames jointly. The encoder attends to temporal motion parallax and relative depth ordering cues to produce a compact set of  $N_{\text{IGT}}$  global geometric tokens. These encode a coarse “3D cognitive map” of the scene: which regions are closer, rough surface orientations, and camera motion type.

**Explicit Geometry Tokens (EGTs)** are produced by running a differentiable SfM module on the input frames to recover a sparse point cloud  $\mathcal{P} = \{(\mathbf{p}_i, \mathbf{n}_i)\}$ . The point cloud is voxelized into a 3D feature grid and encoded by a sparse 3D convolutional encoder into  $N_{\text{EGT}}$  tokens. These encode precise object boundaries, surface normals, and depth discontinuities.

A **3D-aware adapter** fuses IGTs and EGTs with the VLM’s 2D visual tokens using cross-attention with geometry-conditioned masks that spatially align explicit geometric tokens with their 2D image counterparts.

## Technical Formulation

Given video  $V = \{f_1, \dots, f_T\}$ , let  $Z_{2D} \in \mathbb{R}^{N_{2D} \times d}$  be the 2D visual tokens from the VLM’s encoder. The IGT and EGT encoders produce:

$$Z_{\text{IGT}} = \phi_{\text{IG}}(V) \in \mathbb{R}^{N_{\text{IGT}} \times d}$$

$$Z_{\text{EGT}} = \phi_{\text{EG}}(\mathcal{P}(V)) \in \mathbb{R}^{N_{\text{EGT}} \times d}$$

The 3D-aware adapter computes fused tokens:

$$Z_{\text{fused}} = \text{Attn}(Q = Z_{2D}, K = V = [Z_{\text{IGT}}; Z_{\text{EGT}}; Z_{2D}]) + Z_{2D}$$

with a geometry-conditioned attention mask  $\mathbf{M} \in \{0, -\infty\}^{N_{2D} \times (N_{\text{IGT}} + N_{\text{EGT}} + N_{2D})}$  where  $M_{ij} = 0$  if EGT token  $j$  projects within spatial radius  $r$  of 2D token  $i$ , and  $-\infty$  otherwise (no restriction on IGT tokens, which are global). The fused representation  $Z_{\text{fused}}$  concatenated with the text query tokens is passed to the frozen VLM backbone.

## Learning or Inference Procedure

Training proceeds in two stages:

1. **Geometric pretraining:**  $\phi_{\text{IG}}$  and  $\phi_{\text{EG}}$  are pre-trained on 3D video datasets (ScanNet, ARKitScenes) with reconstruction objectives: IGTs supervised to predict surface orientation distributions, EGTs supervised via point cloud completion loss.
2. **End-to-end instruction tuning:** The full model (adapter + pretrained geometry encoders + frozen VLM) is fine-tuned on 3D instruction-following data with a standard language modeling loss.

At inference, only the RGB video is required. The SfM module runs at inference time using COLMAP, adding a modest overhead.

## What the Guarantee Says

VLM-IE3D provides no formal guarantee on 3D accuracy, but the ablation results establish that removing either IGT or EGT degrades performance across all benchmarks—confirming that neither geometry type subsumes the other, and that the dual-geometry design is necessary rather than redundant.

## Experimental Findings

| Benchmark (metric)      | Best Prior | VLM-IE3D |
|-------------------------|------------|----------|
| ScanQA (acc.)           | 46.3       | 52.7     |
| ScanRefer (Acc@0.5)     | 41.2       | 48.9     |
| Scan2Cap (CIDEr@0.5)    | 62.8       | 71.3     |
| EmbodiedScan (mAP@0.25) | 28.4       | 33.1     |

All results are with RGB-only input; prior state-of-the-art methods use ground-truth depth where noted.

## Ablations and Interpretation

**IGT only:** Removes EGT. ScanRefer (grounding) degrades by 6.2 pp, confirming fine-grained local geometry is necessary for precise localization.

**EGT only:** Removes IGT. ScanQA (spatial reasoning) degrades by 4.1 pp, confirming coarse global layout is necessary for scene-level reasoning.

**Naïve concatenation:** Concatenates IGT and EGT without the geometry-conditioned mask. Average performance degrades by 3.8 pp, demonstrating that spatial alignment of explicit geometry tokens is critical.

**SfM quality:** Replacing COLMAP with a faster but noisier SfM variant reduces EmbodiedScan mAP by 2.3 pp, confirming a moderate sensitivity to point cloud quality.

## Reference

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